

Revision checklist

Multi-sensor CR dataset — Nature Scientific Data

All 61 items (Editor + Reviewers 1–5), ranked by difficulty rather than grouped by reviewer. Tap/tick a box or write notes in the margin as you go.

Known bugs to fix first

- ☐ **main.tex** — delete the duplicated 5.31 sentence (appears twice, straddling “motor process is launched last...”).
- ☐ **main.tex** — remove/finish the broken orphan fragment “*The filtering*” tagged 5.36 in Usage Notes / Getting started.
- ☐ **authors response** — fix Comment 5.36: its real Response paragraph is commented out; text is currently sitting under “Changes made” instead.
- ☐ **authors response** — fill in blank Response + Changes for Comment 5.31 (system_clock fix).
- ☐ **authors response** — Comment 5.40 response text is missing the coordinate-frame point — add it.
- ☐ Remember: 5.33’s response claims a joint reprocessing pass with 5.19 — make that true before submitting.
- ☐ Delete the stray duplicate `\end{document}` at the end of the authors-response file.

Trivial (14)

- **5.6 (R5)** Table 1: “Weight” → “Mass” (g, not N). *[drafted]*
- **5.11 (R5)** “minimize” → “reduce” throughout. *[drafted]*
- **5.13 (R5)** Transpose vs. inverse for g_{base} . *[response drafted — explains no change needed]*
- **5.15 (R5)** Merge Data & Code Availability sections. *[drafted]*
- **5.29 (R5)** State FBG delay value (13.4 ms) + sign convention in Methods. *[drafted]*
- **5.31 (R5)** Fix clock name: `system_clock`, not `high_resolution_clock`. *[manuscript drafted, see bugs above]*
- **4.3 (R4)** State CC-BY 4.0 in the paper text. *[drafted]*
- **E3 (Editor)** Remove section numbering.
- **E5 (Editor)** Add data citation (DOI) at start of Data Record.
- **2.3 (R2)** Typo “measuraments” → “measurements”.
- **3.2 (R3)** Figure caption notes (calibration sweeps, zero baselines).
- **4.6 (R4)** Dataset versioning / long-term maintenance note.
- **5.38 (R5)** Rename “Euclidean distance” → “planar distance” (or compute full 3-D).
- **5.16 (R5)** 502 vs. 503 FBG point indices — should read 0–501. *[verified, not drafted]*

Easy (15)

- ☐ **5.36 (R5)** State `filtfilt` is acausal in Usage Notes. *[manuscript drafted, see bugs above]*
- ☐ **5.39 (R5)** Clarify FBG “raw” file already contains Shape CORE-reconstructed data. *[drafted]*
- ☐ **5.40 (R5)** Drop “without further modification”; state actual modifications. *[manuscript drafted, response needs the frame point — see bugs above]*
- ☐ **E1 (Editor)** Reviewer-suggested refs: include only where relevant.
- ☐ **E4 (Editor)** Check Background & Summary for out-of-scope results/conclusions.
- ☐ **1.2 (R1)** Soften “clear physical correspondent in the governing equations” claim.
- ☐ **2.2 (R2)** Tendon–disk contact force sensor limitation statement. *Owner: Chengnan (rationale)*
- ☐ **4.4 (R4)** Clarify 2.2.4 “pose of selected...” sentence. *Cluster with 1.1, 5.7*
- ☐ **4.5 (R4)** Share recording videos if available. *Owner: Greg*
- ☐ **5.5 (R5)** Fix Table 1 description (more than geometrical parameters).
- ☐ **5.7 (R5)** $g_{\text{fix}}/g_{\text{wand}}$ terminology consistency pass. *Cluster with 1.1, 4.4*
- ☐ **5.12 (R5)** Justify 3D-printed pulleys; check roundness/eccentricity. *Owner: Chengnan + Greg*
- ☐ **5.14 (R5)** General prose polish.

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- **5.28 (R5)** Retain OptiTrack validity mask (`disk_k_is_valid`) in processed output. *Owner: Spencer*

 - **5.32 (R5)** State/quantify 4-motor sequential-read timestamp latency. *Owner: Tongjia*

Moderate (21)

- **5.33 (R5)** ATI timestamp end→center correction (-2.5 ms). *[manuscript + code + response drafted; bundled with 5.19, see bugs above]*
- **5.17 (R5)** `mocap_frames.csv` column order — verified [roll,pitch,yaw,x,y,z]. *[not drafted; Spencer sign-off]*
- **5.18 (R5)** “circle” or “Lissajous” logic bug — verified harmless. *[not drafted]*
Owner: Tongjia
- **5.24 (R5)** Quasi-static protocol not reproducible from code — data verified fine. *Owner: Tongjia*
- **1.1 (R1)** Figure/diagram clarity (contact wand, actuator-sensor labeling).
Cluster with 4.4, 5.7
- **2.1 (R2)** Embodied intelligence / morphological computation paragraph.
Owner: Chengnan input
- **3.1 (R3)** Generalization-limitations discussion. *Same E2 tension as 5.8/GVS*
- **4.1 (R4)** Related robots/actuation-architecture paragraph. *Combine with 5.1. Owner: Chengnan input*
- **4.2 (R4)** FBG-vs-MoCap divergence under contact (new analysis). *Owner: Spencer*
- **5.4 (R5)** Quantify contact-force validation (RMSE, bias, max error, correlation).
- **5.20 (R5)** Reconcile contact-processing inconsistencies (15 Hz vs. 30 Hz filter, filenames).
- **5.21 (R5)** `runAllDataCollection.bat` missing Resense/HEX12 process.
Owner: Tongjia / Angela
- **5.22 (R5)** Confirm which OptiTrack calibration procedure generated released data. *Owner: Spencer*
- **5.23 (R5)** Rewrite trajectory description to match discrete-waypoint implementation. *Owner: Tongjia*

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- **5.26 (R5)** Align cross-correlation code with text (exploratory vs. reported).
 - **5.27 (R5)** Reframe FBG–OptiTrack comparison as post-registration consistency check.
 - **5.30 (R5)** Separate FBG delay estimation from delay correction in `process_data.m`.
 - **5.34 (R5)** Reconcile 2 s (text) vs. 3 s (code) pre-motion period. *Owner: Tongjia*
 - **5.35 (R5)** Verify no NaN edge samples after 100 Hz resampling.
 - **5.37 (R5)** Check Mark-10 noise calculation (raw pulley force vs. $\div 2$ tension).
 - **5.41 (R5)** Bundle of small code/label cleanups.

Larger (4)

- **5.1 (R5)** Related-work citations/paragraph — 5 bib entries drafted, paragraph not written. *Combine with 4.1*
- **5.2 (R5)** Figure 2 redesign (contrast, alignment, capitalization).
- **5.3 (R5)** Mounting stiffness / structural eigenfrequency bench test. *Owner: Greg*
- **5.19 (R5)** ATI Fs bug (29–34× overestimate, effective cutoff 0.52 Hz instead of 15 Hz). Code fix drafted; **full reprocessing not yet run**. *Owner: Andrea*

Blocked — needs a decision first (7)

- **E2 (Editor)** Root blocker: Data Descriptors must not include results/conclusions/discussion/analyses. Needs a framing decision.
- **5.8 (R5)** Add a conclusion/summary section. *Blocked on E2*
- **5.9 / 5.10 / 1.3 / 4.7** GVS baseline: release code, run comparison, write up. **Highest leverage — answers 4 reviewers**. *Owner: Andrea. Blocked on E2*
- **5.25 (R5)** Soften temporal-synchronization claims. *Blocked on confirming acquisition PC's Python version*

Open questions for a teammate

- **FBGS “Angle” frame consistency** Shape is rotated into robot frame; Angle/Curvature aren’t. Curvature is frame-invariant (fine), but per Moore & Rogge 2012, Angle may be bend-*direction*. **Needs Spencer** to confirm against Shape CORE definition.
- **Acquisition-machine Python version** Needed for 5.19/5.25 (Windows `time.time()` precision history). *Ask Tongjia*

Duplicate `\label{sec:postprocessing}` — checked, now appears only once. Worth a final grep before submission.

Owner summary

Andrea (self): GVS baseline, ATI reprocessing (5.19+5.33), mocap doc fix, Figure 2, editor items, most Trivial/Easy items.

Tongjia: 5.18, 5.23, 5.24, 5.32, 5.34; optionally 5.21.

Chengnan: 5.12 (rationale), 4.1/5.1 (related work), light input on 2.1.

Spencer: 5.22, 4.2, 5.17 (sign-off), 5.27, 5.28; FBGS Angle frame-consistency question.

Greg: 5.3 (stiffness bench test), 4.5 (videos), physical half of 5.12.